

THE CONSTRUCTION OF THE FTD MATHEMATICS

From a Discrete Ontology to Its Provable Boundary

A MONOGRAPH ON FOUNDATIONAL TERNARY DYNAMICS

*Derive everything the discrete ontology forces — and
rigorously establish the boundary of what it cannot force.*

2 June 2026

SCOPE

This monograph builds FTD’s mathematics from the ground up, in the order a mathematician would build the number systems, for an interdisciplinary and philosophy-of-physics readership. It is a construction and an honest accounting at once: it introduces no new mathematics, and it states every result at exactly the grade the result has earned — theorem, derivation, conjecture, or open problem — and at no grade higher. That discipline is not a hedge; it is the point. It lets the strong results stand as strong and the boundary stand as a genuine finding.

HOW TO READ THIS MONOGRAPH

This is a *construction story*. It builds FTD’s mathematics in the order a mathematician would build the number systems — each object from prior ones, each step independently checkable — and it asks one question at every step: **does the discrete ontology force this, or not?** (where “the discrete ontology” includes the one $\mathbb{Z}[i]$ modelling choice of §I.1). The answer comes with an epistemic tag, and the tag is the point. A reader from philosophy of physics is invited to scrutinize the tags first and the prose second; the tags are where the project’s honesty lives.

The monograph is organized around the project’s single governing goal, stated as two clauses:

Derive everything the discrete ontology forces — and rigorously establish the boundary of what it cannot force.

Both clauses are deliverables. Showing that the discrete ontology *does not* fix a quantity is as much a result of this program as showing that it does; a mapped boundary is not a failure but a finding.

How a tag looks. Throughout, an epistemic tag is set as a small-caps badge whose *rule* — never its colour — carries its tier, so the page reads identically in grayscale. There are four treatments:

THEOREM	FORCED — rigorously established (heavy rule)
PARAMETRIC	CONDITIONAL — calibration / selection (thin rule)
CONJECTURE	CONJECTURAL — evidence, no derivation (dashed rule)
OPEN	BOUNDARY — a mapped wall / open kernel (double rule)

The same rule signatures frame the displayed “marquee” statements and label the construction map of fig. 0.1. The full contract — every tag, its meaning, and what a reviewer should expect of it — is the table of §0.3.

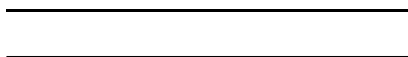
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PART 0

THE SEED



Part 0 lays the foundation the rest of the monograph stands on: the discrete ontology FTD posits, the standard of “construction” the document holds itself to, the tag system the reader must internalize, and the two-clause roadmap that organizes everything that follows. It contains **no physics theorem and no new mathematics**; it is setup.

0.1 THE DISCRETE ONTOLOGY

FTD posits a discrete substrate, fixed by five postulates. Each is an **AXIOM**: a structural commitment that defines the model and is **not** claimed to be derived from anything more primitive. A reader should accept them as the definition of the game, then ask what follows.

AXIOM

1. **Discrete space.** The substrate is a three-dimensional cubic lattice with **undefined boundary**: at every specified position, the axis-adjacent sites exist. This is deliberately *not* a completed-infinity totality $\mathbb{Z}^3$ taken as a single finished object; it is a lattice with no defined edge and no claim of completeness.
2. **Discrete time.** Evolution proceeds in discrete ticks; there is no continuous time parameter at the substrate level.
3. **Ternary states.** Each site holds a state $s \in \{-1, 0, +1\}$, the three values being the only states a site may take.
4. **Local causality.** Influence is local: each site interacts only within its 26-neighbour Moore neighbourhood, and information propagates at most one lattice unit per tick.
5. **Determinism.** The update rule is deterministic: the configuration at tick $t + 1$ is fixed by the configuration at tick t .

The two-layer ontology **AXIOM**. Riding on the lattice are two fields with distinct ontological roles:

- a **flux field** $J \in \mathbb{R}^3$, a continuous vector field at each site, encoding potential energy density. It is *dispositional*: it represents what the substrate is poised to do, not what it has manifested.
- a **state field** $s \in \{-1, 0, +1\}$, the discrete ternary value above. It is *actual*: it represents manifestation, what the substrate has resolved to.

The dispositional/actual distinction is structural to the framework and recurs throughout; the flux field carries the continuous content, the state field carries the discrete resolution.

Why the undefined boundary is load-bearing. The choice of an undefined-boundary lattice over a completed-infinity $\mathbb{Z}^3$ is not cosmetic; it narrows the framework’s admissible toolkit and is enforced corpus-wide. Concretely, statements of the form “in the limit $L \rightarrow \infty \dots$ ” are **not well-posed** under this ontology without an explicit ε – L restatement (e.g. “for every $\varepsilon > 0$ there exists L such that $P(L)$ is within ε of its continuum counterpart”). Global integrals over “all sites,” completed thermodynamic limits, and path integrals over “all configurations” are likewise not primitive moves; arbitrarily large *finite* computations, explicitly bracketed, are. This commitment will matter precisely at the boundary in Part II, where the question “does the ontology force a particular limiting value?” turns on what limits the ontology is even entitled to take.

0.2 WHAT A CONSTRUCTION IS

This monograph holds itself to a specific standard, and naming it up front is part of the contract with the reader.

A construction, in the sense used here, is the kind of thing the number systems are: $\mathbb{N} \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{R} \rightarrow \mathbb{C}$. Each object is built *explicitly* from objects already in hand ($\mathbb{Z}$ as equivalence classes of pairs of naturals; $\mathbb{Q}$ as equivalence classes of pairs of integers; $\mathbb{R}$ as cuts or Cauchy sequences; $\mathbb{C}$ as $\mathbb{R}^2$), and each construction step is *independently checkable* by a reader who accepts the prior stage. No object appears by fiat; none is assumed because it would be convenient. FTD’s mathematical core is presented in exactly this spirit: each result built from prior results and the axioms, with a stated proof or chain the reader can audit. The ordering is fixed: mathematical primitives $\rightarrow$ invariant structure $\rightarrow$ admissible readouts $\rightarrow$ operational physics, with physical language entering *only* at the last layer. The constructive Parts of this monograph (I and II) live in the first two layers; the physics *match* is withheld until Part III, and Part II names the physical target only to state precisely what is *not* forced.

Three categories must be kept strictly apart, because conflating them is the characteristic way a program of this kind overclaims:

- **A derivation** (tagged **THEOREM** or **DERIVED**) is an *explicit chain* from the axioms or prior theorems that the document itself reproduces. The reader checks the chain. **DERIVED** is the honest grade when

the chain is explicit but carries a non-trivial assumption; **THEOREM** when it does not.

- **A parametric insertion** (tagged **PARAMETRIC**) is a standard physics formula with FTD’s constants substituted in. The numbers may fit beautifully, but the *mechanism is borrowed* from outside the framework: a calibration input, not a derivation.
- **A match** (at its strongest tagged **STRONGLY MOTIVATED CONJECTURE**) is a numerical coincidence between an FTD object and a physical quantity, possibly backed by uniqueness scans or sub-ppm agreement, but with **no derivation chain**. A match is evidence, not a derivation.

For every load-bearing claim, this monograph states which of the three it is.

0.3 THE EPISTEMIC CONTRACT

The following tag system is the instrument the whole program is governed by. A reader should internalize it before proceeding; every substantive claim sentence in this monograph carries one of these tags, and the tag (not the surrounding prose) is the claim’s actual epistemic status.

TAG	MEANING	REVIEWER EXPECTATION
AXIOM	Structural postulate; not derivable.	Accept as model definition.
THEOREM	Rigorously proven from axioms (or from named classical theorems with explicit citation).	Check the proof.
DERIVED	Established by an explicit chain the document reproduces; weaker than a theorem when the chain carries a non-trivial assumption.	Check the chain; flag any smuggled axiom.
SELECTION	Argued from consistency or naturalness; not uniquely proven.	Critique the argument.

TAG	MEANING	REVIEWER EXPECTATION
STRONGLY MOTIVATED CONJECTURE	A conjecture with substantial structural and/or empirical evidence (uniqueness scan, multi-route convergence, sub-ppm match) but no derivation chain.	Critique the evidence; expect an explicit Bayes / uniqueness / look-elsewhere argument.
CONJECTURE	Proposed interpretation requiring validation; weaker than the above (no structural-uniqueness backing).	Demand evidence.
PARAMETRIC	A standard physics formula filled with FTD constants; numbers fit, mechanism borrowed.	Treat as calibration input, not output.
CLOSED NEGATIVE	A hypothesis tested and falsified; preserved for provenance to prevent re-attempt.	Confirm the closure; cite it to prevent a zombie re-emergence.
SYNTHESIS	Cross-document integration of existing claims at their existing tags; not a new theorem.	Verify the component claims; check that nothing was silently promoted.

The reading rule. The tags form a ladder, and the ladder has a direction:

*Promotion up the ladder requires explicit justification.
Demotion is free. Ambiguous cases default down.*

A claim may be moved *up* (e.g. from `CONJECTURE` to `DERIVED`) only by exhibiting the chain or proof that warrants it. A claim may be moved *down* at any time, by anyone, for any defensible reason, with no ceremony. And where the correct tag is genuinely unclear, the convention is to assign the *weaker* tag until the stronger one is earned. This asymmetry is the safeguard against the program talking itself into believing its own conjectures.

Every result appears at the grade it has earned, and at no grade higher. (Four further tags appear in the body where they are needed: `MEASURED` for an honest measurement, `IMPOSED` for an imposed

choice, OPEN for an open problem, and FOUNDATIONAL OBSTRUCTION for a foundational obstruction.)

0.4 THE TWO CLAUSES: ROADMAP

The program's governing goal has two clauses, and the architecture of this monograph is simply those two clauses made into sections, plus a quarantined bridge to physics.

Clause 1: Derive everything the discrete ontology forces.

Clause 2: Rigorously establish the boundary of what it cannot force.

Clause 1 → Part I, the Constructive Reach. Part I exhibits what the five postulates *force* once their order-4 lattice symmetry is read as $\mathbb{Z}[i]$ (the single modelling AXIOM of §I.1): an algebraic spine of theorem-grade results: the constant $G^* = \Gamma(1/4)/\Gamma(3/4)$, the master quadratic $P(x) = x^2 - 16 G^{*2} x + 16 G^{*3}$ and its roots, the arithmetic of the lemniscatic CM curve, and the finite combinatorics of the Moore neighbourhood. The canonical accounting is **nine numbered results, of which seven are theorem-grade** and two are honestly tiered below theorem grade. These are the framework's firm ground, independent of any physics interpretation, and Part I states each at its canonical tag.

Clause 2 → Part II, the Boundary, and this is the climax. Part II asks the sharp negative question: granted everything in Part I, does the discrete ontology *force* the electromagnetic coupling α ? The honest answer, mapped across multiple independent attack routes, is **no, not without an additional binding law**. The boundary takes a concrete and memorable form: the reading that would fix α requires the substrate to *natively realize* the quantity $\sqrt{G^*(4G^* - 1)}$ (the discrete limit of a square root) as an operator-assembly the lattice produces on its own, and across every FTD-native route examined the lattice does not produce it; the determinant grading and the operator assembly $(\text{Tr } W, \det W) = (16 G^{*2}, 16 G^{*3})$ are an imposed selection, not a forced consequence. The boundary itself is built from theorem-grade parts around one precisely-stated open kernel: a ruled-out geometric realization, a closed-scope restriction, and a route-invariant reduction (the same kernel reached in four different dialects, because $\mathbb{Q}(G^*)$ is the Galois-fixed field of the master quadratic's $\mathbb{Z}/2$). Part II's

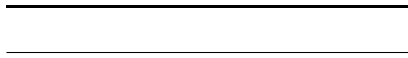
deliverable is therefore a *mapped wall*: theorem-grade masonry around one clearly-marked gap, exactly the kind of boundary Clause 2 calls for.

The bridge → Part III, quarantined. The numerical match that motivates the whole enterprise ($x_+ = 137.0361714582\dots$, agreeing with $1/\alpha$ to 1.26 ppm) is real and is the strongest structural evidence the framework holds (a unique dual-matcher across millions of candidate polynomials over a basket FTD did not design). But it is a *match*: it carries the tag `[STRONGLY MOTIVATED CONJECTURE]`, not a derivation, and Part III presents it as such, alongside the `[PARAMETRIC]` physics insertions.

The central honest headline, stated up front. The discrete ontology, **once its order-4 lattice symmetry is read as $\mathbb{Z}[i]$ (the single modelling `[AXIOM]` isolated in §I.1)**, forces a rich algebraic spine (seven theorem-grade results), and it **provably does not force the electromagnetic coupling α without an additional binding law**. Both halves are the deliverable. A program that only proved the first half would be a curiosity; a program that mapped the second half honestly is doing the harder and more useful thing: drawing, in both directions, the line of what discreteness reaches. The constant $G^* \approx 2.95868$ that anchors the spine is, throughout, the lemniscatic ratio $\Gamma(1/4)/\Gamma(3/4)$; it is **not** the Bernoulli/Gauss lemniscate constant $\varpi \approx 2.62206$, a distinct number with which it is never to be conflated.

PART I

THE CONSTRUCTIVE REACH



Part I answers Clause 1: **what does the discrete ontology force?** It exhibits the algebraic spine (the theorem-grade results the substrate produces) in the order a mathematician would build them, each object from prior ones, each step independently checkable. The construction runs $i \rightarrow \mathbb{Z}[i] \rightarrow G^* \rightarrow \text{master quadratic} \rightarrow \text{spine}$. It carries **no physics identification**: the famous numerical match $x_+ = 1/\alpha$ is a *match*, deferred to Part III, and Part I says so at the one place it is tempting to overstep.

The spine is **nine numbered results: seven are theorem-grade, and two are honestly tiered below theorem grade** (the coefficient-16 identity, whose structural necessity is conjectural, and the Phase-J ultralocality, which is a theorem only at $L = 2$). Section I.4 closes by re-stating that count.

I.1 THE SEED: i AND $\mathbb{Z}[i]$

The construction begins with one structural choice and one group-theory fact, and it is worth separating them cleanly, because the honesty of everything downstream depends on not confusing the two.

The structural choice. The substrate is the cubic lattice of Part 0. Its coordinate planes carry an order-4 rotational symmetry: a 90° rotation of the (x, y) plane sends $(1, 0) \mapsto (0, 1) \mapsto (-1, 0) \mapsto (0, -1) \mapsto (1, 0)$, a cycle of length 4. FTD's seed move is to *read this order-4 planar symmetry as the arithmetic of the Gaussian integers* $\mathbb{Z}[i] = \{a + bi : a, b \in \mathbb{Z}\}$, with the 90° rotation realized as multiplication by i . That the lattice's quarter-turn *should be read through* $\mathbb{Z}[i]$ rather than through some other order-4 structure is a modelling decision, not a theorem, the discrete-ontology analogue of choosing to build $\mathbb{C}$ as $\mathbb{R}[i]$: natural, motivated, and a *choice*. We mark it an **AXIOM** and flag it as such, so that no later theorem silently inherits the status of “forced” from a step that was actually “chosen.”

AXIOM

The group-theory facts. Once $\mathbb{Z}[i]$ is the substrate's arithmetic, the following are ordinary, checkable theorems of algebra, owing nothing further to FTD:

THEOREM

- The unit group of $\mathbb{Z}[i]$ is $\mathbb{Z}[i]^\times = \{1, i, -1, -i\} \cong \mathbb{Z}/4$, of order $|\mathbb{Z}[i]^\times| = 4$. (*Proof:* $a + bi$ is a unit iff its norm $a^2 + b^2 = 1$, whose only integer solutions are $(\pm 1, 0)$ and $(0, \pm 1)$.)

- $\mathbb{Z}[i]$ is a Euclidean domain, hence a PID and a UFD; its primes split into the three Gauss classes: the ramified prime $1+i$ (norm 2), the split rational primes $p \equiv 1 \pmod{4}$, and the inert rational primes $p \equiv 3 \pmod{4}$ (Fermat's two-square theorem).

The single integer this seed deposits into the rest of the construction is $4 = |\mathbb{Z}[i]^\times|$. It reappears, squared, as the coefficient 16 of the master quadratic (§I.3), and the split/inert residue classes mod 4 reappear as the arithmetic content of one of the four G^* constructions (§I.2). Hold the distinction firmly: 4 is a *theorem* about $\mathbb{Z}[i]$; that the *lattice* is entitled to $\mathbb{Z}[i]$ in the first place is the axiom.

I.2 G^* IN FOUR REPRESENTATIONS OF DIFFERING PROVENANCE

The bridge constant is

THEOREM

$$G^* = \frac{\Gamma(1/4)}{\Gamma(3/4)} = 2.95867511919\dots$$

What earns G^* its place at the centre of the spine is that it admits four *representations* of sharply differing character: a Γ -ratio, a lattice Green's-function period, a regularized determinant ratio, and a finite product. These are not four independent witnesses: three share the quarter-integer data, and the Watson value is itself a $\Gamma(1/4)$ -period, so all four are representations of one and the same CM period of the lemniscatic curve. What is striking is that a single period wears four such different faces (arithmetic, lattice-analytic, spectral, and finitary), one of which (Watson's body-centred-cubic integral) is of genuinely independent provenance. Each is shown where it is cheap and cited where it is deep.

Route 1: the Γ -ratio. Directly from the definition and the Euler reflection formula,

$$\Gamma(1/4) \Gamma(3/4) = \frac{\pi}{\sin(\pi/4)} = \pi\sqrt{2},$$

multiplying $G^* = \Gamma(1/4)/\Gamma(3/4)$ by $\Gamma(3/4)^2$ and substituting gives the two standard closed forms

$$G^* = \frac{\Gamma(1/4)^2}{\pi\sqrt{2}} = \frac{2\varpi}{\sqrt{\pi}},$$

where $\varpi = \frac{\Gamma(1/4)^2}{2\sqrt{2\pi}} \approx 2.62205755$ is the Bernoulli/Gauss lemniscate constant. A three-line proof, verified to 50 digits.

Route 2: the Watson BCC period, conditional on Watson 1939. Let W_3 be the body-centred-cubic Watson integral

$$W_3 = \frac{1}{\pi^3} \int_0^\pi \int_0^\pi \int_0^\pi \frac{dk_1 dk_2 dk_3}{1 - \cos k_1 \cos k_2 \cos k_3}.$$

Watson's 1939 closed-form evaluation [12], consolidated by Glasser–Zucker [5], gives

$$W_3 = \frac{G^{*2}}{2\pi}.$$

The BCC sub-lattice is one of the three polyhedral layers of the substrate's Moore neighbourhood, so this is G^* arising from a *lattice* quantity, a Green's-function period of an entirely different character from Route 1's Γ -ratio. "Conditional on Watson 1939" means the rigour equals that of the cited classical evaluation, not that it rests on a conjecture. Verified to 100 digits in PARI.

Route 3: the $\det_\zeta$ quarter-conjugacy bridge. Let J be the conjugacy operator with $J^2 = -I$ (multiplication by i ; the same Z_4 seed of §I.1). A wavefunction on S^1 with the quarter-twisted boundary condition $\psi(\varphi + 2\pi) = J\psi(\varphi)$, evolved by the natural quarter-period operator $-i d/d\varphi$, carries its two J -eigensectors on the shifted spectra $D_{1/4} = \{n + 1/4\}_{n \geq 0}$ and $D_{3/4} = \{n + 3/4\}_{n \geq 0}$ (the choice of that operator is a modelling choice, like the seed, not something forced). Lerch's formula for the ζ -regularized determinant of an arithmetic progression,

$$\det_\zeta \{n + a\}_{n \geq 0} = \frac{\sqrt{2\pi}}{\Gamma(a)},$$

yields the determinant ratio

$$\frac{\det_\zeta D_{3/4}}{\det_\zeta D_{1/4}} = \frac{\sqrt{2\pi}/\Gamma(3/4)}{\sqrt{2\pi}/\Gamma(1/4)} = \frac{\Gamma(1/4)}{\Gamma(3/4)} = G^*,$$

the $\sqrt{2\pi}$ cancelling. The arithmetic is the reason this route belongs to the same story as the seed: $4 \cdot D_{1/4} = \{n \equiv 1 \pmod{4}\}$ and $4 \cdot D_{3/4} = \{n \equiv 3 \pmod{4}\}$ are exactly the two non-trivial residue classes mod 4, and, restricted to primes, the split and inert prime classes of $\mathbb{Z}[i]$ (§I.1). G^* is the regularized asymmetry between them.

Route 4: the finite- N attractor. Define the finite product

$$G_N^* := (N+1)^{-1/2} \prod_{n=0}^N \frac{n+3/4}{n+1/4}.$$

Every G_N^* is computable in $O(N)$ rational operations with no transcendental ever invoked ($G_0^* = 3$, $G_1^* = 21/(5\sqrt{2}) \approx 2.9698$, $G_5^* \approx 2.95995$, $G_{20}^* \approx 2.95878$). A Stirling expansion of $\Gamma(N+7/4)/\Gamma(N+5/4) \sim (N+1)^{1/2}$ gives

$$G_N^* = G^* + \frac{C}{N^2} + O(N^{-3}), \quad \text{so } G_N^* \rightarrow G^* \text{ at rate } O(1/N^2)$$

(empirical $C \approx 0.046$). This route does double duty. It is a fourth construction of G^* , and it **discharges the Part-0 ε - L obligation** for G^* : the undefined-boundary ontology forbids treating $G^* = \lim_{N \rightarrow \infty} G_N^*$ as a primitive completed limit, and Route 4 supplies exactly the finitary restatement the ontology demands: G_N^* is defined for every finite N and approaches G^* at the stated rate. G^* is therefore reachable by a convergent finite computation, no completed infinity required.

G^* is the **lemniscatic ratio**, not ϖ . $G^* \approx 2.95868$ is the ratio $\Gamma(1/4)/\Gamma(3/4)$. It is **not** the Bernoulli/Gauss lemniscate constant $\varpi \approx 2.62206$, a distinct number related by $G^* = 2\varpi/\sqrt{\pi}$. Informal usage sometimes calls both “the lemniscate constant”; here they are never conflated, and the distinction is load-bearing in the most literal sense. The master quadratic of §I.3 produces $x_+ = 137.0361714582 \dots$ **only** at $G^* = 2.95868$; substituting $\varpi = 2.62206$ into the same polynomial gives $x_+ \approx 107.3$, nowhere near $1/\alpha$.

I.3 THE MASTER QUADRATIC

With G^* in hand, the construction’s central polynomial is purely algebraic.

THE MASTER QUADRATIC THEOREM

Define

$$P(x) = x^2 - 16 G^{*2} x + 16 G^{*3}.$$

Its discriminant factors cleanly,

$$\begin{aligned}\Delta &= (16 G^{*2})^2 - 4 \cdot 16 G^{*3} \\ &= 256 G^{*4} - 64 G^{*3} = 64 G^{*3} (4G^* - 1),\end{aligned}$$

which is **positive** because $G^* > 1/4$ (indeed $G^* \approx 2.96$), so both roots are real:

$$\begin{aligned}x_{\pm} &= 8 G^{*2} \pm \sqrt{64 G^{*4} - 16 G^{*3}} \\ &= 8 G^{*2} \pm 4 G^* \sqrt{4 G^{*2} - G^*}.\end{aligned}$$

Numerically $x_+ = 137.0361714582 \dots$ and $x_- = 3.0239639163 \dots$

This is the quadratic formula applied to a polynomial whose coefficients are integer multiples of powers of G^* ; nothing more is involved. Verified numerically across 54 independent checks.

The coefficient-16 soft spot, flagged honestly. The prefactor 16 has a tempting arithmetic reading: for the lemniscatic CM elliptic curve $E : y^2 = x^3 - x$, the automorphism group over $\overline{\mathbb{Q}}$ is $\text{Aut}(E) = \{\pm 1, \pm i\} \cong \mathbb{Z}/4$ (the unit group of $\mathbb{Z}[i]$ again), so $|\text{Aut}(E)| = 4$ and $|\text{Aut}(E)|^2 = 16$. That $|\text{Aut}(E)|^2 = 16$ is a **THEOREM**. But that the master quadratic's prefactor is *forced* to equal $|\text{Aut}(E)|^2$ (that two objects equal 16 *for a structural reason*) is not proven: it is a **CONJECTURE** on the structural necessity, the framework's softest mathematical spot. A partial unification exists (a tower-level reading supplies a reason for $k = 4$, hence $2^k = 16$, via $|\mathbb{Z}[i]^\times|^2 = 16$), but a partial unification is not a forcing theorem, and the honest grade stays **CONJECTURE** on necessity.

No physics here. $x_+ = 137.036 \dots$ is, in *Part I*, a root of a polynomial, nothing more. The proximity to $1/\alpha$ is the motivation for the whole enterprise, but the identification $x_+ \leftrightarrow 1/\alpha$ is a **STRONGLY MOTIVATED CONJECTURE** deferred in full to **Part III**; it is not asserted as forced anywhere in the constructive Parts. The smaller root $x_- \approx 3.024$ is a pure algebraic artifact, pinned by the Vieta relation $x_- = 16G^{*3}/x_+$ the instant x_+ is, and it carries no physics. The historical identification $x_- \leftrightarrow N_c$ is **retired**; FTD's $N_c = 3$

is independently sourced from the Moore-neighbourhood topology, owing nothing to the polynomial's smaller root.

I.4 THE SPINE BUILT ON TOP

On the foundation of G^* (§I.2) and the master quadratic (§I.3), a further family of theorem-grade results stands. Each is stated at its exact grade and condition; short proofs are shown, deep external dependencies are cited.

Harmonic-invariant tower. For each integer $k \geq 3$, the $(1+i)$ -tower master quadratic is $M_k(x) = x^2 - 2^k G^{*k-2} x + 2^k G^{*k-1}$ (the $k = 4$ instance is §I.3's polynomial). Writing the normalized roots $y_{\pm} := x_{\pm}/G^*$, at every level $k \geq 3$,

THEOREM

$$\frac{1}{y_+} + \frac{1}{y_-} = 1.$$

The proof is a three-line Vieta computation: $1/x_+ + 1/x_- = (x_+ + x_-)/(x_+x_-) = (2^k G^{*k-2})/(2^k G^{*k-1}) = 1/G^*$; multiplying through by G^* gives $1/y_+ + 1/y_- = 1$. Verified to 50 digits.

CM-curve uniqueness, under the trivial-multiplier criterion.

THEOREM

Among all imaginary quadratic fields $K = \mathbb{Q}(\sqrt{-d})$, the field $\mathbb{Q}(i)$ ($d = 1$, fundamental discriminant -4) is the **unique** one satisfying $|\mu_K| = |\text{disc}(K)|$: for $\mathbb{Q}(i)$, $|\mu_K| = 4$ and $|\text{disc}(K)| = 4$; for every other imaginary quadratic field the two differ. (*Proof sketch: $|\mu_K| \in \{2, 4, 6\}$ with 4 only at $d = 1$ and 6 only at $d = 3$; $|\text{disc}(K)| \geq 3$ always, $= 3$ only at $d = 3$, $= 4$ only at $d = 1$; case-checking the three unit-group orders leaves $d = 1$ as the sole coincidence.*) The trivial-multiplier criterion is load-bearing: a “match” here requires the natural root to equal the target directly. Under a looser rational-multiplier criterion, 20 additional non-canonical matches exist in the tested grid, and that reading is a **SELECTION**, not a theorem. The proof is a complete case analysis over all imaginary quadratic fields; the $d \leq 200$ computation is a redundant cross-check.

$\mathbb{Q}(G^*)$ is π -free, conditional on Chudnovsky 1976. $\mathbb{Q}(G^*) \cap \mathbb{Q}(\pi) = \mathbb{Q}$: the field generated by G^* shares no algebraic content with $\mathbb{Q}(\pi)$ beyond the rationals. The proof reduces a hypothetical polynomial relation between G^* and π (via the identity $G^* \cdot \pi \cdot \sqrt{2} = \Gamma(1/4)^2$) to a polynomial relation between π and $\Gamma(1/4)$, which Chudnovsky's

THEOREM

1976 algebraic-independence theorem [3] forbids. “Conditional” here means the rigour equals that of this published, foundational transcendence result (consolidated in Waldschmidt [11]), not that it rests on a conjecture.

Tower-discriminant transcendence , conditional on Schneider–Chudnovsky [3, 8]. For the tower of the harmonic invariant, the discriminant factors as $\text{disc}(M_k) = 2^{k+2} G^{*k-1} A_k$, $A_k := 2^{k-2} G^{*k-3} - 1$. Then A_k is rational at $k = 3$ ($A_3 = 1$) and **transcendental over $\mathbb{Q}$** (i.e. $A_k \notin \overline{\mathbb{Q}}$) at every $k \geq 4$ ($A_4 = 4G^* - 1$, etc.): a non-rational polynomial with rational coefficients in the transcendental G^* takes transcendental values.

THEOREM

BCC complex structure. The 8 BCC corners under 90° rotation form two orbits of size 4, and, as a module over the cyclic group $C_4 = \langle 90^\circ \text{ rotation} \rangle \cong \mathbb{Z}/4$,

THEOREM

$$\mathbb{Z}[\text{BCC}] \otimes \mathbb{Q} = V_{\text{triv}}^2 \oplus V_{\text{sign}}^2 \oplus V_{\text{complex}}^2,$$

where V_{complex} carries a natural $\mathbb{Z}[i]$ -module structure $\cong \mathbb{Z}[i]^2$. Paired with it is a clean no-go: there is **no** injective homomorphism $\mathbb{Z}[i]^\times \rightarrow O_h^{\text{ab}}$, since $\mathbb{Z}[i]^\times \cong \mathbb{Z}/4$ has an order-4 element but $O_h^{\text{ab}} \cong \mathbb{Z}/2 \times \mathbb{Z}/2$ (Klein four) does not, a one-line group-order argument. Together these say the substrate’s $\mathbb{Z}[i]$ structure lives genuinely in the BCC layer but does **not** extend to a global lattice-symmetry embedding. Verified in exact rationals.

Lemniscatic L -value , conditional on Rubin 1991. The central L -value of the lemniscatic curve $E_{\text{lemn}} : y^2 = x^3 - x$ (Cremona 32.a3) has the clean closed form

THEOREM

$$L(E_{\text{lemn}}, 1) = \frac{\varpi}{4} = \frac{G^* \sqrt{\pi}}{8} \approx 0.6555143885,$$

via the full BSD formula for the CM rank-0 case [7]. (An earlier $\varpi/2$ came from a BSD-convention double-count; the corrected factor is $\varpi/4$, verified to 27 digits against the curve’s entry in LMFDB [9].)

Three further results, conditional on the classical theorems named. The character χ_{-4} four-level unification — χ_{-4} on $(\mathbb{Z}/4\mathbb{Z})^\times$ generating the entire $G^*/G_{\mathbb{G}}$ identity algebra through four functorial projections (lattice, Chowla–Selberg, Hecke, Dirichlet) — holds conditional on Deligne’s period conjecture [4] in the CM case, itself proved

THEOREM

unconditionally (Blasius/Anderson/Shimura). The η -tower across the nine class-number-one imaginary-quadratic fields,

$$|\eta(\tau_K)|^{2w_K} = \frac{G_K^{w_K}}{(2\pi|d_K|)^{w_K/2}},$$

holds conditional on Chowla–Selberg [2], unifying the Heegner near-integer phenomenon as a χ -projection. Here G_K is the field-specific Chowla–Selberg period (it equals G^* only for the Gaussian field $d = 1$), so the nine fields share the *form* of the identity, not the constant. The $\text{Sym}^2 \oplus \text{Sym}^3$ structure singles out the construction’s pair. Among the period polynomials $x^2 - 16G^{*a}x + 16G^{*b}$ with $a < b$ positive integers and positive discriminant (which requires only $b < 2a + \log 4/\log G^* \approx 2a + 1.28$, i.e. $b \leq 2a + 1$ for integers), the pair $(a, b) = (2, 3)$ is distinguished by the *parity of G^* beneath its discriminant’s radical*: $\Delta_{(2,3)} = 64G^{*3}(4G^* - 1)$ carries an odd power of G^* , so $\sqrt{\Delta} = 8G^* \sqrt{G^*(4G^* - 1)}$ introduces the generator $\sqrt{G^*(4G^* - 1)}$ (the very square root that walls Part II), whereas the neighbouring pairs carry an even power ((1, 3) gives $\sqrt{\Delta} \propto \sqrt{4 - G^*}$; (1, 2) and (2, 4) have $b = 2a$). Equivalently, the squarefree radicand of Δ retains a bare factor of G^* exactly on the line $b = 2a - 1$ (the pairs (2, 3), (3, 5), (4, 7), $\dots$, each adjoining the wall field $\mathbb{Q}(G^*)(\sqrt{G^*(4G^* - 1)})$), of which (2, 3) is the minimal member with $a < b$. That much is elementary algebra; the stronger reading (that (2, 3) is the *unique* natural target) is a SELECTION, argued from naturalness, not asserted here as a theorem.

Phase-G geometric Coulomb. The remaining theorem-grade node is a lattice fact. On the cubic lattice the static potential between two manifested sites solves the lattice Poisson equation, and its solution is, at every finite L , exactly the periodic-lattice Poisson Green’s function G_L . The emergent inverse-square Coulomb tail is therefore lattice geometry, carrying no fine-structure content of its own: the lattice coupling reads as a zero-parameter geometric object, $\alpha_r(r, L) = 2r G_L(r)$, not a free constant. This is a finite- L identity, established directly, with no continuum limit required.

The two honestly-tiered results. Two of the nine numbered results sit *below* theorem grade and are stated as such:

- **Coefficient-16:** the value-level identity $16 = |\text{Aut}(E)|^2$ is true, but its structural necessity is a CONJECTURE, as set out in §I.3.

CONJECTURE

- **Phase-J partition-function ultralocality:** a THEOREM AT $L = 2$ ONLY.

THEOREM AT $L = 2$ ONLY

On a 2^3 lattice the classical Euclidean action depends on the state field solely through $\sum_i s_i^2$ (the manifested-site count) and is invariant under spatial permutation of charge placement; this is proved by explicit construction. The general- L extension is **disconfirmed**: the $L = 2$ ultralocality is a Nyquist-mode counting degeneracy (every non-zero momentum has $\sin(k_i) = 0$), not a structural property, and at $L \geq 3$ the action does depend on spatial placement. The $L = 2$ restriction is stated plainly and is the spine's honest claim.

The canonical accounting. That completes the spine: **nine numbered results, seven theorem-grade** (G^* identity, master quadratic, CM uniqueness, Watson identity, Phase-G geometric Coulomb, harmonic-invariant tower, $\mathbb{Q}(G^*)$ π -freeness conditional on Chudnovsky) **plus two tiered below theorem grade** (coefficient-16 and Phase-J $L = 2$). This is the discrete ontology's firm ground (its answer to Clause 1), and it stands independent of any physics interpretation. The match to physics is a separate matter, quarantined in Part III.

I.5 THE CONSTRUCTION MAP

The spine is not a list but a dependency structure: a directed acyclic graph rooted in the seed. The axiom reading of the lattice's quarter-turn as $\mathbb{Z}[i]$ (§I.1) deposits the integer 4; G^* is constructed four ways from quarter-integer data (§I.2); the master quadratic $P(x) = x^2 - 16 G^{*2} x + 16 G^{*3}$ is assembled from G^* and $16 = 4^2$ (§I.3); and the spine theorems (harmonic tower, CM uniqueness, $\mathbb{Q}(G^*)$ π -freeness, the BCC complex structure, the L -value and its companions) hang off G^* and the quadratic in turn (§I.4). Reading the arrows backward is the audit trail: every theorem-grade node traces, through explicit and checkable steps, to the one structural choice at the root, and to nothing else.

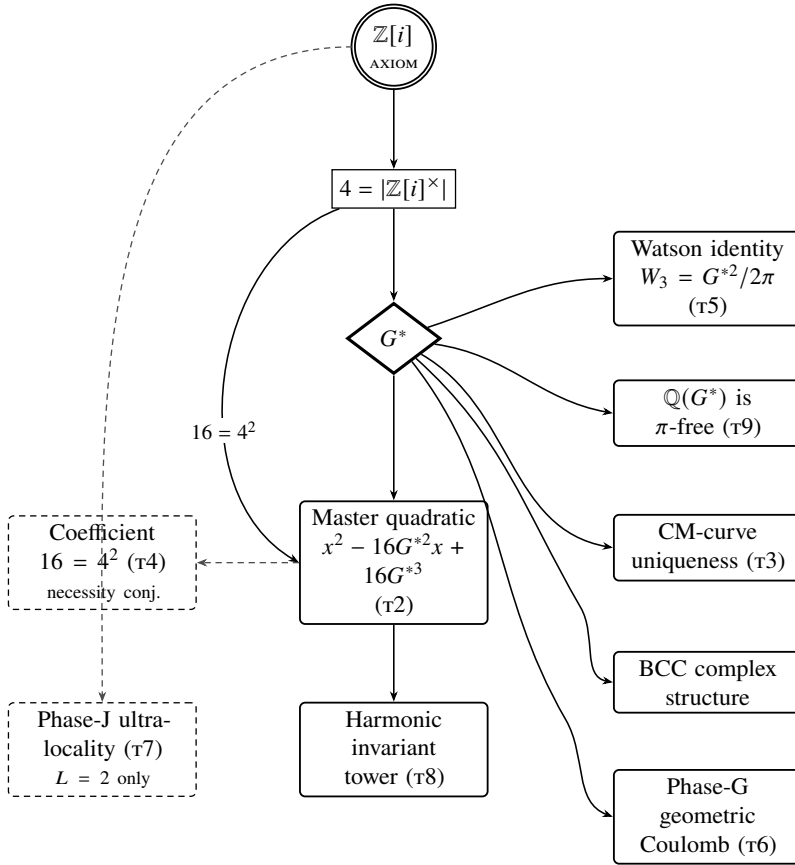
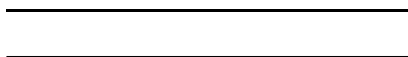


Figure 0.1. The construction DAG. Shape and rule encode epistemic status: the double-ringed circle is the lone AXIOM ($\mathbb{Z}[i]$), the heavy diamond is the hub (G^*), solid boxes are theorem-grade, and dashed boxes are tiered below theorem grade (the coefficient-16 necessity, conjectured; Phase-J, a theorem only at $L = 2$). The chain runs $i \rightarrow \mathbb{Z}[i] \rightarrow 4 \rightarrow G^* \rightarrow$ master quadratic ($16 = 4^2$) $\rightarrow$ spine; reading the arrows backward traces every theorem-grade node to the one structural choice at the root.

PART II

THE BOUNDARY



Part II answers Clause 2: **what does the discrete ontology provably not force?** It is the monograph’s climax, and the section where the program’s honesty is most directly on the line, because the quantity at stake is the one whose numerical match ($x_+ = 137.036 \dots$, deferred to Part III) motivates the whole enterprise. The temptation to call that match a derivation is exactly what this Part refuses. What it delivers instead is a *mapped wall*: a route-invariant boundary around the electromagnetic coupling α , built from theorem-grade masonry around one precisely-located, honestly-marked OPEN gap.

State the shape of the result before its content. The headline — $\mathfrak{F}$ (the five postulates plus the spine) does not force α — is not a single theorem. It is a route-invariant boundary assembled from three theorem-grade components plus one irreducible OPEN kernel; the overall no-go is a STRONGLY MOTIVATED CONJECTURE, and the obstruction it sharpens (the framework’s central foundational obstruction) remains a FOUNDATIONAL OBSTRUCTION. The honest one-line summary is the project’s own: **α is dynamical, not structural.** It is never “ α is derived,” and it is never “ α is proven underivable” in any absolute sense; §II.6 states exactly what it is.

FOUNDATIONAL OBSTRUCTION

II.1 THE READOUT PROBLEM

Part I built the master quadratic $P(x) = x^2 - 16 G^{*2} x + 16 G^{*3}$ as a pure object of the algebraic spine (§I.3): its coefficients are integer multiples of powers of G^* , its roots $x_+ = 137.036 \dots$ and $x_- = 3.024 \dots$ are an exercise in the quadratic formula, and the whole of it is a theorem. To turn that polynomial toward physics, one more thing is needed, and naming it sharply is the entire content of this Part.

The readout move. Read the quadratic as the characteristic polynomial of a 2×2 *readout operator* W , an admissible operational functional on the substrate whose dominant eigenvalue is to be the measured inverse coupling. For its characteristic polynomial to be P , the operator must have

$$(\text{Tr } W, \det W) = (16 G^{*2}, 16 G^{*3}).$$

This is the **operator assembly**. The question that decides whether α is derived or merely matched is exactly:

THE READOUT PROBLEM OPEN

Does $\mathfrak{F}$ (the five postulates plus the algebraic spine plus the O_h representation theory of the BCC corner module) *force* the assembly $(\text{Tr}, \det) = (16G^{*2}, 16G^{*3})$, or is that assembly an *imposed selection*?

This is the readout-structure criterion, and the contract a closure must satisfy is fixed in advance and deliberately strict: an admissible readout must be stated *before* any physical value is checked, and a closure fails immediately if it inserts physical α , a CODATA value, g_c , or x_+ , or uses the framework's transfer matrix $M_N(t)$, which is *defined* to have the master quadratic as its characteristic polynomial, hence circular and banned. The missing object is not another fit; it is an operational readout rule. Part II determines whether the ontology supplies one.

Why this is the sharp form of the obstruction. The central foundational obstruction of the framework's α program comes down to this one structural question, because everything *around* it is already settled in Part I. The constant G^* is forced (four ways, §I.2); the coefficient $16 = |\mathbb{Z}[i]^\times|^2$ is a theorem at the value level (§I.3); the polynomial's roots are a theorem. If the *assembly* were forced too, α would be DERIVED. So the entire weight of the derive-or-match verdict rests on one hinge: whether $\mathfrak{F}$ produces W .

II.2 THE FOUR FTD-NATIVE ROUTES

A boundary cannot be argued from absence, from merely failing to find an assembly. The canonical attack instead ran **four FTD-native routes of differing character** (later shown, in §II.3, to be four dialects of one underlying obligation, so “0 of 4” is one structural fact reached four ways, not four independent failures). Each was first *force-attempted* (build the strongest honest forward chain to the assembly, G^* kept symbolic, all banned moves excluded) and then *adversarially refuted* by a second pass hunting for any smuggled step:

The verdict: 0 of 4 force the assembly. Every force-pass self-reported a gap; every refute-pass upheld boundary; the set of cleanly-forced routes is empty. The honest reading partitions cleanly into what the ontology *does* force and what it does *not*:

STRONGLY MOTIVATED
CONJECTURE NO-GO

ROUTE	FTD-NATIVE PRINCIPLE ATTEMPTED
jtwist	the J -twisted ζ -regularized determinant operator (the candidate clean odd source)
bcc	the BCC body-diagonal transfer / response operator (triple-cosine Watson)
cm	the lemniscatic-CM arithmetic of $E : y^2 = x^3 - x$ (CM by $\mathbb{Z}[i]$, $\text{Aut} = \mu_4$)
novel	a forced variational principle / period-ring (Hodge) valuation / K -theory co-realizability

- **What IS forward-forced** (DERIVED), agreed by every route with nothing smuggled:
 - the **even trace** $16G^{*2}$: $16 = |\mu_4|^2$ for $E : y^2 = x^3 - x$; $G^{*2} = 2\pi \cdot G_{\text{BCC}}(0)$, the Watson BCC self-energy;
 - the **existence of a clean FTD-native odd source**: the J -twisted determinant ratio $\det_{\zeta}(D_{3/4})/\det_{\zeta}(D_{1/4}) = \Gamma(1/4)/\Gamma(3/4) = G^*$ (degree 1; the $\sqrt{2\pi}$ cancels, so there is no forbidden $\sqrt{\pi}$ prefactor). This is *strictly stronger than trace-only*: it means $16G^{*3} = 16G^{*2} \cdot G^*$ is **assemblable**, genuinely lifting the bare-parity no-go.
- **What is NOT forced** (OPEN):
 - the **operator assembly itself**: that the *same* 2×2 readout carrying $\text{Tr} = 16G^{*2}$ also has $\det = 16G^{*3}$. For a 2×2 operator, trace and determinant are **independent invariants**; fixing the trace leaves the determinant free. The odd scalar G^* exists, but nothing forces the *gluing*: neither that the two invariants belong to one operator, nor that the odd scalar lands in the determinant slot rather than anywhere else. This residual is precisely the imposed Vieta target of the readout-structure criterion.

The differing character of the four channels is what makes “0/4” load-bearing rather than anecdotal; and §II.3 turns the fact that they reach one obligation into a theorem. The overall no-go is a [STRONGLY MOTIVATED CONJECTURE], not a theorem: the routes are adversarial refutation *attempts*; none constructed the forcing assembly, and none proved one *cannot* exist. Calling the boundary a theorem on this

evidence would overreach. It is a boundary-mapping deliverable in the exact sense of Clause 2, at honest conjecture grade.

A representative route, made concrete. The boundary shows itself cleanly in the modular dialect. At the self-dual point $\tau = i$ (the fixed point of the modular S -transformation $\tau \mapsto -1/\tau$), the torus partition function's modular geometry forces $E_6(i) = 0$ and $E_4(i) \propto G^{*4}$, so the available invariants are *even* powers of the period G^* (G^{*2} from the Green's-function variance, G^{*4} from E_4). The required determinant $16G^{*3}$ is an *odd* power, which no theta or modular form on the torus generates without inserting an external scale. The verdict is exactly UNDERDETERMINED: this route supplies the even trace and cannot force the odd determinant slot, one instance of the route-invariant pattern the next section names.

II.3 ROUTE-INVARIANCE AND THE THEOREM-GRADE COMPONENTS

The deeper attack asked whether the residual gap could be closed by a sharper instrument; and in failing to close it, established three genuinely theorem-grade facts plus the reason all four routes land in the same place. These are the boundary's *masonry*: each is a theorem, and together they hem the open kernel of §II.4 into a single, precisely-located gap.

(a) The flip is ruled out. A *flip* would be a substrate-native operator that genuinely *forces* $\det = 16G^{*3}$, flipping the verdict to “ α derived.” The only geometric candidate is the $C_3(\langle 111 \rangle)$ three-plane $\det_\zeta$ product (a D_6 -symmetric object built from three cyclically-permuted planes, each contributing one factor of G^*). It is **excluded** from the rank-2 readout in two steps:

THEOREM

- a definite complex structure J with $J^2 = -I$ (required by the trace via the BCC $V_{\text{complex}} = \mathbb{Z}[i]^2$ structure) needs $\text{mult}_O(E) = 0$ (the character table of O shows no O -symmetric 2-dimensional subspace of the 8-corner module), which forces O_h to break to **one C_4 axis**, so $C_3(\langle 111 \rangle) \notin \text{Stab}$;
- the D_6 three-plane product requires $C_3(\langle 111 \rangle) \in \text{Stab}$;
- but a 4-fold face axis and a 3-fold body-diagonal axis generate the whole octahedral rotation group, $\langle C_4, C_3 \rangle = O$ (order 24), so

demanding both restores unbroken $O_h \Rightarrow$ no localized charge $\Rightarrow$
no $V_{\text{complex}} \Rightarrow$ no readout.

The two requirements are mutually exclusive from a single preparation. Every 2×2 that *does* realize $(16G^{*2}, 16G^{*3})$ (e.g. the companion form $\begin{bmatrix} 0 & -16G^{*3} \\ 1 & 16G^{*2} \end{bmatrix}$) has its determinant entry *hand-placed*, which is exactly the banned gluing: a witness for the cheap argument, not a derivation.

(b) The equivariant restriction closes its own scope. No $C_3(\langle 111 \rangle)$ -equivariant rank-2 *restriction* of the three-plane source carries $(\text{Tr}, \det) = (16G^{*2}, 16G^{*3})$. On the C_3 -adapted basis $\mathbb{R}^3 = R \oplus C$, the unique rank-2 C_3 -invariant subspace is the C -plane, whose C_3 -commutant is $\{xI + yK\} \cong \mathbb{C}$ (Schur). The master-quadratic roots are real and distinct ($\text{disc} = 64G^{*3}(4G^* - 1) > 0$), and a C_3 -equivariant C -plane operator has real-distinct spectrum *iff it is non-real*, which invokes the ambient scalar i , supplied only by breaking O_h to one C_4 axis, whence $\langle C_4, C_3 \rangle = O$ and there is no readout. The chain is therefore **reality $\Rightarrow$ scalar- $i \Rightarrow C_4 \Rightarrow O$** : the *same* wall as the rest of the program, reached from the restriction side, not an independent conjugacy fact. The same mechanism holds dimension-free: any C_3 -equivariant linear reduction of the three-plane object factors through the C_3 -fixed diagonal (where C_3 acts as the identity), collapsing the determinant grading $G^{*3} \rightarrow G^{*1}$; the rank-2 compressions of $G^* \cdot I_3$ have determinant G^{*2} at most, never G^{*3} . This closes the restriction's scope; the surviving non- C_3 -invariant and det-by-fiat cases are exactly the open kernel.

THEOREM

(c) The reduction is route-invariant. All four attacks reduce to the *same* obligation in four dialects, and this is not a coincidence; it follows from one clean field-theoretic fact:

THEOREM

$\mathbb{Q}(G^*)$ is the Galois-fixed field of the master quadratic's $\mathbb{Z}/2$.

The quadratic's discriminant is $\Delta = 64G^{*3}(4G^* - 1)$, so its splitting field over $\mathbb{Q}(G^*)$ is $\mathbb{Q}(G^*)(\sqrt{G^*(4G^* - 1)})/\mathbb{Q}(G^*)$, with Galois group $\mathbb{Z}/2$ swapping $x_+ \leftrightarrow x_-$. Every forward-forced *symmetric* FTD-native datum (the Watson trace $16G^{*2}$, the $\det_\zeta$ ratio G^* , the Chowla–Selberg periods) lives in the σ -fixed subfield $\mathbb{Q}(G^*)$ and is therefore **provably blind to which root is $1/\alpha$** . The concrete witness: the family $\det = 16G^{*2} \cdot G^{*k}$ for $k = 0, 1, 2, 3$ gives dominant roots 139.05 / 137.04 / 130.68 / 105.76, *all* of them $\mathfrak{F}$ -consistent, and

nothing in $\mathfrak{F}$ selects $k = 1$. The regularized traces $\zeta(-1, 1/4) = \zeta(-1, 3/4) = 1/96$ are rational, so the trace carries zero G^* ; and at rank 2 the regularized determinant is the ordinary product $x_+ \cdot x_- = 16G^{*3}$, so hitting the assembly is a *chosen* Vieta target, not a forced one.

These three components are why the boundary is a *wall* and not a mere absence of progress: the obvious forcing candidate is dead (a), the equivariant restriction is closed within its scope (b), and any remaining route is provably the *same* route in disguise (c). What is left is one gap, and §II.4 states it exactly.

II.4 THE SQUARE-ROOT WALL

The three theorem-grade components hem the obstruction into a single irreducible obligation. Stated once, plainly:

THE BINDING OBLIGATION OPEN

Prove (or refute) that no substrate-native operator construction can bind the degree-1, C_3 -agnostic odd scalar G^* (the J -twisted $\det_\zeta$ ratio) into the determinant slot of the *same* 2×2 readout that carries the definite complex structure i , with the exponent fixed at exactly **1** by the substrate rather than chosen.

The basis-free fact is the primary one: trace and determinant are independent invariants of a rank-2 operator over $\mathbb{Q}(G^*)$, so the assembly W is a free choice. The square-root form below is the cleanest single-coordinate dialect of that one fact, a basis-representative of a basis-free obstruction, not a coordinate-bound claim (the determinant's parity is itself basis-dependent: a rescaling migrates the odd content into the trace). In that dialect: to fix α , the substrate must **natively realize**

$$\sqrt{G^*(4G^* - 1)},$$

the **squarefree generator** of the quadratic extension $\mathbb{Q}(G^*)(\sqrt{\Delta})/\mathbb{Q}(G^*)$; since $\Delta = 64G^{*3}(4G^* - 1)$ we have $\sqrt{\Delta} = 8G^* \cdot \sqrt{G^*(4G^* - 1)}$, the irrational quantity that *first distinguishes* x_+ from x_- . Every *symmetric* datum lives in $\mathbb{Q}(G^*)$ and cannot tell the two roots apart (§II.3(c)); the square root is exactly the ingredient that breaks the $\mathbb{Z}/2$ symmetry and picks a root. This

is the discrete-limit-of-the-square-root wall: the precise place the construction halts. The substrate forces every *symmetric* ingredient of the quadratic; it does not, on present evidence, produce the *one antisymmetric beable* (the square root) that would select the physical branch.

Why it is OPEN, and not closeable today. The obligation is a **universal negative** over substrate-native operator constructions, and it is not yet a well-posed proposition, because $\mathfrak{F}$ does not yet pin down the class it quantifies over. Provisionally, take that class to be the 2×2 operators on $V_{\text{complex}} = \mathbb{Z}[i]^2$ generated by the substrate-native data (the BCC corner module, the C_4 -winding preparation, and the $\det_\zeta$ functor) closed under direct sum and restriction to a stabilizer (the regularized determinant furnishing the characteristic invariants, not itself a closure operation). Whether this is the *right* closure, whether a larger one (tensor products, non-self-adjoint generators) would change the verdict, is itself part of the open work, not settled here. Until that calculus is fixed precisely, “no substrate-native operator binds the odd scalar at exponent 1” quantifies over an undefined domain, and one cannot prove a universal over a class one has not finitely specified. Two senses of OPEN are worth separating: an open *conjecture* is a well-posed proposition awaiting proof; an open *desideratum* is a property not yet expressible as a proposition because its quantifier-domain is unspecified. The binding obligation is currently the latter; fixing the operator calculus would convert it to the former. **Crucially, the conditional theorem of §II.5 is insulated from this:** it rests only on trace and determinant being independent rank-2 invariants over $\mathbb{Q}(G^*)$ (which is well-posed and proved), so the boundary’s logic does not wait on the operator calculus.

The one non-axiomatic exit is currently shut. The alternative to axiomatizing the calculus is to *measure* the binding directly on the substrate. That measurement has been run, and it returned a CLOSED-NEGATIVE: a null robust to measurement precision, **0 macroscopic cluster fissions across 2000 seeds** in the tested regime, from a protocol that would have detected fissions above its sensitivity floor and saw none. One should not read “impossible” into a finite null; but neither route, the axiomatic (no calculus) nor the empirical (a null result so far), is presently open, and this is the single remaining obligation of the entire α program.

II.5 THE CONDITIONAL THEOREM

What *is* provable, rigorously and unconditionally, is the conditional. It is the strongest theorem-grade statement the boundary supports, and the precise sense in which “ α is not forced” is earned rather than asserted:

THE CONDITIONAL THEOREM THEOREM

Nothing that $\mathfrak{F}$ forces selects the operator assembly $(\text{Tr}, \det) = (16G^{*2}, 16G^{*3})$. Every datum the spine forward-forces is *symmetric*; it lies in the fixed field $\mathbb{Q}(G^*)$ of the master quadratic’s $\mathbb{Z}/2$, and is therefore blind to which root is $1/\alpha$. With the trace $16G^{*2}$ already forward-forced, only the determinant slot is free: the assembly is one of an infinite family of determinant values over that fixed trace, all equally compatible with everything $\mathfrak{F}$ forces. To single it out (equivalently, to realize a beable in $\mathbb{Q}(G^*)(\sqrt{G^*(4G^* - 1)}) \setminus \mathbb{Q}(G^*)$) demands a substrate-native binding law W beyond $\mathfrak{F}$ that fixes the determinant’s odd- G^* exponent at exactly **1** from a single stabilizer.

The force of this conditional is carried by the §II.3 components, above all the Galois-fixed-field fact, which is what makes “ $\mathfrak{F}$ does not select the assembly” a theorem and not a restatement of ignorance: the two sides are exhibited explicitly, as realizations, not merely asserted.

- **Adopting W is coherent**, the master quadratic is the explicit realization: once the assembly is posited it is internally consistent.
- **Declining W is equally coherent**: $\det = 16G^{*4}$ (dominant root ≈ 130.68) and $\det = G^*$ (dominant root ≈ 140.04) are explicit alternative assemblies that everything the spine forces equally permits.

Because both are coherent, no binding law is implied by what $\mathfrak{F}$ forces; and *that*, not any failed search, is the content of “the ontology does not force the assembly.” W is, in the framework’s bookkeeping, a **6th-postulate-class input**: added to FTD’s axioms *and* shown substrate-native, it would convert $x_+ = 1/\alpha$ from a STRONGLY MOTIVATED CONJECTURE to DERIVED, MODULO W . Whether any FTD-native W exists is exactly the binding obligation of §II.4, which remains OPEN. The conditional

itself is theorem-grade; the *unconditional* no-go (“no FTD-native W can ever exist”) is the universal negative, which stays open, and is why the overall boundary is a STRONGLY MOTIVATED CONJECTURE.

II.6 CONCLUSION: α IS DYNAMICAL, NOT STRUCTURAL

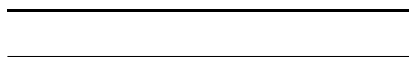
The deliverable of Part II is a **mapped wall**. In the project’s own phrasing: **the discrete ontology forces the menu** (the even trace $16G^{*2}$ from the Watson integral, and the existence of a clean odd source G^* from the J -twisted ζ -determinant), **but it does not force the dish**, the assembly of those ingredients into one readout operator W . So α ’s value rides on an independent convention, not on the postulates. The assembly is unforced on present evidence: that contrast is DERIVED; the reading it licenses, α **is dynamical, not structural**, is the STRONGLY MOTIVATED CONJECTURE that summarizes it, not a proof that no future binding law exists. The contrast with N_c is exact and instructive: $N_c = 3$ *is* structural, forced from O_h and topology by four independent routes with no operator-assembly choice required; the value falls out of the symmetry. α has no such forcing: with the trace forward-forced, everything the ontology fixes is still consistent with infinitely many determinant values over that one trace, the same epistemic status the engine assigns its coupling g_c .

Accordingly, the central obstruction remains a FOUNDATIONAL OBSTRUCTION, and $x_+ = 1/\alpha$ remains a STRONGLY MOTIVATED CONJECTURE; nothing in this Part promotes either. The wall is built of theorem-grade masonry (the ruled-out flip, the closed-scope restriction, the route-invariant reduction, and the conditional theorem) around one clearly-marked open gap: the square root that would break the root-swapping symmetry. The surviving exits are explicit and few: a 6th-postulate-class input W that *forces* the assembly, or a fresh discrete-native measurement (and the one run so far returned a CLOSED NEGATIVE).

This is exactly what Clause 2 of the governing goal asks for. “*Rigorously establish the boundary of what the ontology cannot force*” is not a consolation prize for failing to derive α ; it is a deliverable in its own right. Part II draws that line where the construction actually halts: at the discrete limit of a square root. A program that only exhibited the spine of Part I would be a curiosity; one that maps this wall, in both directions and at honest grades, is doing the thing the goal was set to do.

PART III

THE BRIDGE



Part III is the **physics quarantine**. Parts I and II did their work in mathematics, naming the physical target in Part II only to state precisely what is *not* forced: an algebraic spine the ontology *forces* (Clause 1), and a route-invariant boundary it provably does *not* force (Clause 2). This Part introduces the one ingredient withheld until now (the **numerical match to physics**) and states it, and the broader physics layer around it, at the exact grade each piece carries. It **derives nothing**; it states, with the grades foregrounded.

The discipline here is stricter than anywhere else in the monograph, because physics is where overclaim is easiest and most consequential. The rule for this Part, stated once and enforced throughout: **no sentence attaches a [THEOREM] or a [DERIVED] to α , to a particle mass, to a mass formula, or to a gauge ratio**. The α match is a [STRONGLY MOTIVATED CONJECTURE]; the rest of the construction layer is [PARAMETRIC], with two lepton-anchored mass formulas one tier higher, stated at that tier and no higher.

III.1 THE EMPIRICAL MATCH

Here is the match that motivates the entire enterprise. The master quadratic's larger root (Part I, §I.3) agrees with the reciprocal of the electromagnetic fine-structure constant [10]:

THE EMPIRICAL MATCH [STRONGLY MOTIVATED CONJECTURE]

$$\begin{aligned} x_+ &= 137.0361714582\dots, \\ \alpha^{-1} &= 137.035999177(21) \quad (\text{CODATA 2022}), \\ \frac{|x_+ - \alpha^{-1}|}{\alpha^{-1}} &= 1.26 \times 10^{-6} = \mathbf{1.26 \text{ ppm}}. \end{aligned}$$

A discrete ontology, built only to be internally coherent, throws up a polynomial one of whose roots lands on $1/\alpha$ to better than two parts in a million. That is the seed of the whole program.

And it is a match, not a derivation. The identification $x_+ \leftrightarrow 1/\alpha$ is the framework's single load-bearing physics claim, and it is a [STRONGLY MOTIVATED CONJECTURE]. Part II established *why* it cannot be more: the discrete ontology does not force the operator assembly that would read x_+ as the physical coupling, so α is **dynamical, not structural**

[STRONGLY MOTIVATED
CONJECTURE]

(§II.6). The honest sentence is the conjunction of both halves: *the number matches to 1.26 ppm, and the ontology does not derive that it should.*

III.2 THE STRUCTURAL-UNIQUENESS EVIDENCE (AND ITS HONEST LIMIT)

A 1.26-ppm coincidence invites the obvious objection: with enough constants and enough integers, *some* simple expression will land near any target. The framework’s answer is a structural-uniqueness argument, and it is the strongest evidence the program holds.

The adversarial look-elsewhere scan. The scan asked: among all degree-2 polynomials of the master quadratic’s natural form over a basket of 18 physical constants *that FTD did not design*, how many match the framework’s two targets as well as the master quadratic does? The answer is **zero non- G^* dual-matchers across 2.65 million degree-2 polynomials**, the master quadratic is the **unique** dual-matcher in the search space, **rank 1 by $\sim 130\times$** over the nearest competitor. Against a random-coincidence null this is $\sim 4 \times 10^5 : 1$ in favour of the structural reading. Two supporting nulls sharpen the picture: the **Eisenstein-family null** (the Eisenstein-integer multiplier family contributes zero dual-matchers) and the **Chowla–Selberg $h \geq 2$ null** (no higher-class-number CM field reproduces the match through the Γ -product analogue). This is a genuine measured fact (MEASURED), and it is the cleanest reason to take the match seriously rather than dismiss it as numerology.

The honest limit. The evidence is strong, and it is *evidence*, not a derivation chain, which is exactly what the STRONGLY MOTIVATED CONJECTURE grade records. The $\sim 4 \times 10^5 : 1$ figure is computed *within a declared natural family*; the full look-elsewhere correction across *all* of the framework’s degrees of freedom (every constant it might have admitted, every polynomial form it might have selected) is not something a single scan can close. By the project’s own Bayesian convention a figure above $10^6 : 1$ is “decisive”; $\sim 4 \times 10^5 : 1$ is deliberately short of that. The evidence materially raises the posterior that the match is not coincidence, and it remains evidence, not proof.

III.3 RETIRED AND CLOSED-NEGATIVE

A construction story is incomplete without its dead ends: preserving them is what stops a closed route from re-emerging as a fresh “discovery.” Three categories of negative result quarantine the physics claim from its discredited neighbours.

The smaller root carries no physics: $x_- \leftrightarrow N_c$ **is retired.** The smaller root $x_- \approx 3.024$ was once identified with the QCD colour number $N_c = 3$ (a 0.80% match). That identification is **retired**: as Part I noted (§1.3), x_- is a pure algebraic artifact, fixed by the Vieta relation $x_- = 16G^{*3}/x_+$ the instant x_+ is, not an independent physical quantity. FTD’s $N_c = 3$ is independently sourced (from the Moore-neighbourhood topology, by four convergent routes) and owes nothing to the polynomial’s smaller root.

The x_- search closed negative. That x_- carries no physics was not merely asserted; it was *tested*. A pre-registered, hash-locked search evaluated $x_- = G^*/(1 - \alpha G^*) \approx 3.02396$ against **exactly 25 pre-specified Standard Model observables** from CODATA 2022 / PDG 2024 [6, 10], under strict algebraic, structural, and dual-match uniqueness filters. **All 25 fired the algebraic-miss falsifier; zero survived.** x_- has no Standard Model correspondent: it is a pure coordinate/chirality artifact of the quadratic (CLOSED NEGATIVE).

The α -derivation routes are all closed. Every attempt to *derive* α from FTD’s dynamics (rather than match it) has been run and falsified, and each is preserved to prevent re-attempt: **R1** (transverse stiffness), **R2** (source-current normalization), **R3** (two-sector response eigenvalue), **R4** (projected Dirac matter), the **Z-factor reading**, **RG-running**, **algebraic combinations**, $1/\sqrt{d}$, **Langevin-equipartition**, and the **monomial scans**, all (CLOSED NEGATIVE). The list is not a record of failure but a map: it is the empirical content of Part II’s verdict that α is dynamical, traced route by route. The single non-derivation route that remains live is the readout/observable-selection class, whose boundary Part II mapped.

III.4 WHAT WOULD CLOSE THE GAP

The gap is not a mystery; it is a precisely-located obligation, and the honest statement of the research frontier is the statement of what

would discharge it. Part II named the two exits, and they are the only two on present evidence:

1. **A 6th-postulate-class binding law W .** An additional structural input (outside the five postulates) that *forces* the operator assembly $(\text{Tr}, \det) = (16G^{*2}, 16G^{*3})$, equivalently that natively realizes the antisymmetric beable $\sqrt{G^*(4G^* - 1)}$ which selects the physical root (the binding obligation of §II.4, OPEN). Added to FTD's axiom list *and* shown substrate-native, it would convert $x_+ = 1/\alpha$ from a STRONGLY MOTIVATED CONJECTURE to DERIVED, MODULO W . Whether any FTD-native W exists is the open kernel of the entire α program.
2. **A fresh discrete-native measurement.** A measurement of the binding directly on the substrate, bypassing imported continuum machinery. This exit is currently shut: the one run so far returned a CLOSED NEGATIVE, a null robust to measurement precision, 0 macroscopic cluster fissions across 2000 seeds in the tested regime (§II.4).

These are the live frontier, stated as obligations rather than promises. Neither is presently open, and that is exactly why $x_+ = 1/\alpha$ stands where it does. The monograph claims no imminent closure; it claims to have located the gap precisely enough that either exit, if it lands, will be recognizable as a closure rather than another match.

III.5 THE HONEST PHYSICS SCOPE

The α match is the framework's sharpest physics claim. The broader "construction layer" (the mass spectrum, the gauge ratios, the gravity and quantum-mechanics-emergence layers) is weaker, and graded as such.

The construction layer is parametric. The bulk of FTD's Standard Model coverage consists of **standard physics formulas filled with FTD constants**: the six quark masses, the ~ 42 meson and ~ 48 baryon masses, the ~ 22 decay rates and widths, the precision-QED observables ($g-2$, the Lamb shift), and gauge ratios such as $\sin^2 \theta_W = \frac{3}{13}$ (3.5% from the on-shell value) and $\alpha_s = \frac{7}{59}$ (0.63% error). The functional form is borrowed from QED, chiral perturbation theory, Regge phenomenology, or the renormalization group, and FTD supplies the integers or couplings. These are PARAMETRIC insertions:

PARAMETRIC

calibration cross-checks against known physics, not derivations. The wider mixing-angle sector is moreover competitor-dense: several ratios face multiple rational competitors at the same tolerance, and $\sin^2 \theta_{13}$ is effectively a mis-prediction, which is exactly why these are cross-checks, not independent evidence. Of roughly **162** catalogued Standard Model quantities, about **23** are **DERIVED**, **129** are **PARAMETRIC**, and **10** are **IMPOSED** or **SELECTION**. That distribution (a small genuinely-derived core and a large parametric cross-check layer) is the honest shape of FTD's physics reach.

Two lepton-anchored mass formulas sit one tier higher. The electron mass

STRONGLY MOTIVATED
CONJECTURE

$$m_e = m_P \sqrt{2\pi} \frac{16}{3} \alpha^{11} \quad (0.19\% \text{ error})$$

and the proton–electron ratio

$$\frac{m_P}{m_e} = \frac{N_{\text{eff}}}{\alpha} + N_{\text{base}} N_{\text{eff}} + N_c \quad (173 \text{ ppm error})$$

are each a **STRONGLY MOTIVATED CONJECTURE**, not merely **PARAMETRIC**: they combine FTD's Moore-neighbourhood integers $\{N_{\text{base}}, N_{\text{eff}}, N_c\} = \{4, 13, 3\}$ with α in structurally-motivated combinations, which is harder to dismiss as a bare rational fit. But the prefactors and exponents are motivated rather than dynamically derived, so they are stated at that grade and no higher.

The gravity layer. The identification of the engine constant $G_N = 0.01$ with the physical Newton constant is a **CLOSED NEGATIVE**. What the substrate derivation actually yields is the gravitational fine-structure ratio for one electron,

$$\alpha_G(e, e) = \left(\frac{m_e}{m_P}\right)^2 \approx 1.745 \times 10^{-45} \quad (0.38\% \text{ match}),$$

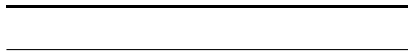
and that result carries one flagged interpretive step (the clock hypothesis), so it is not advanced here as a clean derivation of G .

Dimensionless vs. dimensional: where the falsifiable spine lives.

One structural fact organizes all of the above. FTD's dimensional predictions (any value in MeV, seconds, or metres) are **calibration-conditional**: they pass through exactly two theorem-enforced anchors, $a_{\text{phys}} \equiv \ell_P$ (length) and $K_B = m_e$ (mass), and are no firmer than

those declared calibrations (m_e above consumes the $K_B = m_e$ anchor, which is why it is not a free dimensional prediction). The **dimensionless** predictions (α , the lepton mass *ratios* m_μ/m_e and m_τ/m_e , the mixing angles) require no calibration and constitute the calibration-independent falsifiable spine. The dimensionless ratios are where FTD stakes a falsifiable claim; the dimensional values inherit the status of their calibration; and the broad mass/gauge layer is a PARAMETRIC cross-check, not independent evidence.

Coda



The construction is complete. It is worth saying, in plain language and for the reader who came to this document from philosophy of physics rather than from lattice field theory, what the finished map shows, and what it honestly does not.

The two findings, stated together. A finite, discrete ontology — five postulates, no completed infinity, no primitive continuum, together with the single $\mathbb{Z}[i]$ reading of its order-4 symmetry (§I.1) — **forces** a rich theorem-grade algebraic spine: the constant $G^* = \Gamma(1/4)/\Gamma(3/4)$ (in four representations of one CM period), the master quadratic and its roots, the arithmetic of the lemniscatic CM curve, the harmonic-invariant tower, the π -freeness of $\mathbb{Q}(G^*)$: **seven theorem-grade results** in all (Part I). And the *same* ontology **provably does not force** the electromagnetic coupling α without an additional binding law: the boundary is route-invariant, built from theorem-grade masonry around one precisely-located open kernel, and its honest verdict is that α is *dynamical, not structural* (Part II). **Both halves are the deliverable.** This is the point that is easiest to miss and most important to hold: the second finding is not the first finding's failure. A mapped boundary (*this* is what the ontology reaches, and *this* is the precise place it stops) is a result in its own right, a different and non-trivial kind of result, one easy to mistake for a failure. A program that exhibited only the spine would be a curiosity; one that maps the wall, in both directions and at honest tags, is doing the thing the goal was set to do.

The least-wrong self-assessment. Stated without flattery: FTD is a **philosophy-of-mathematics project with a rigorous algebraic core and suggestive (not derived) physics connections**. The algebraic core is genuine: the seven theorem-grade results survive skeptical mathematical review, are independently checkable, and stand independent of any physical interpretation. The physics connection is real and is honestly weaker than the mathematics: the α match is a [STRONGLY MOTIVATED CONJECTURE] backed by strong structural-uniqueness evidence and no derivation chain; the broader mass-and-gauge layer is [PARAMETRIC]. FTD is therefore **not** a competitor to QED or general relativity on their empirical terms; it does not out-predict them, and it does not claim to. Its contribution is of a different kind: a demonstration, carried out at theorem grade where it can be and at honest conjecture grade where it cannot, of exactly how far a discrete ontology reaches toward physics on its own, and exactly where it needs an additional input it does not contain.

Where it touches the open questions. The program brushes against two genuinely open foundational questions, and it touches each at that question's honest status, claiming no resolution. On **the measurement problem**, FTD's quantum-mechanics-emergence layer (the Born rule, collapse-as-projection) is $\boxed{\text{SELECTION}}/\boxed{\text{OPEN}}$, not a solution: the load-bearing step "probability = normalized energy density" is unproven, and the framework records it as unproven. On **the status of mathematical structure in physics** (whether the unreasonable effectiveness of mathematics reflects something forced or something chosen) FTD offers an unusually crisp data point precisely *because* it draws the line: here is a case where one can exhibit, with proofs, that a discrete mathematical ontology forces a specific rich structure, and can exhibit, also with proofs, that the same ontology does not force a specific physical coupling. The effectiveness is real on the forced side and absent on the unforced side, and the boundary between them is mapped rather than asserted. That is the most the project claims, and it claims it at the tags the claim can bear.

The two-clause goal, closed. The value of this construction is not a derivation of physics from discreteness; that, Part II shows, the ontology does not supply. The value is the **map**: a faithful drawing of the line of what a finite discrete ontology reaches, **honest in both directions**: every theorem proved where the ontology forces, every boundary marked where it does not, and no tag promoted to make the picture prettier than it is. The constant at the centre of it all, $G^* \approx 2.95868$, is the lemniscatic ratio $\Gamma(1/4)/\Gamma(3/4)$ throughout, never the distinct lemniscate constant $\varpi \approx 2.62206$. That is the construction, and that is its honest reach.

Canonical sources & references

- [1] R. Bringhurst. *The Elements of Typographic Style*. 3rd ed. Hartley & Marks, 2004.
- [2] S. Chowla and A. Selberg. “On Epstein’s Zeta-Function”. In: *Journal für die reine und angewandte Mathematik* 227 (1967), pp. 86–110.
- [3] G. V. Chudnovsky. *Contributions to the Theory of Transcendental Numbers*. Vol. 19. Mathematical Surveys and Monographs. Contains the algebraic independence of π and $\Gamma(1/4)$ (announced 1976). American Mathematical Society, 1984.
- [4] P. Deligne. “Valeurs de fonctions L et périodes d’intégrales”. In: *Automorphic Forms, Representations and L-Functions*. Vol. 33, part 2. Proceedings of Symposia in Pure Mathematics. American Mathematical Society, 1979, pp. 313–346.
- [5] M. L. Glasser and I. J. Zucker. “Lattice Sums”. In: *Theoretical Chemistry: Advances and Perspectives*. Ed. by H. Eyring and D. Henderson. Vol. 5. Academic Press, 1980, pp. 67–139.
- [6] S. Navas et al. “Review of Particle Physics”. In: *Physical Review D* 110.3 (2024), p. 030001. doi: 10.1103/PhysRevD.110.030001.
- [7] K. Rubin. “The “main conjectures” of Iwasawa theory for imaginary quadratic fields”. In: *Inventiones Mathematicae* 103.1 (1991), pp. 25–68. doi: 10.1007/BF01239508.
- [8] T. Schneider. “Arithmetische Untersuchungen elliptischer Integrale”. In: *Mathematische Annalen* 113 (1937), pp. 1–13. doi: 10.1007/BF01571618.

- [9] The LMFDB Collaboration. *The L-functions and Modular Forms Database, elliptic curve 32.a3*. <https://www.lmfdb.org/EllipticCurve/Q/32/a/3>. Accessed 2026. 2024.
- [10] E. Tiesinga, P. J. Mohr, D. B. Newell, and B. N. Taylor. “CODATA Recommended Values of the Fundamental Physical Constants: 2022”. In: *Reviews of Modern Physics* 97.2 (2025), p. 025002. DOI: 10.1103/RevModPhys.97.025002.
- [11] M. Waldschmidt. *Diophantine Approximation on Linear Algebraic Groups*. Vol. 326. Grundlehren der mathematischen Wissenschaften. Springer, 2000.
- [12] G. N. Watson. “Three Triple Integrals”. In: *The Quarterly Journal of Mathematics* os-10.1 (1939), pp. 266–276. DOI: 10.1093/qmath/os-10.1.266.

Colophon

This monograph was set in Times-like NEWTX with L^AT_EX on the `ftdmonograph` class. Its design is deliberately monochrome: epistemic tier is carried by rule weight and style (heavy, thin, dashed, double), never by colour, so the argument reads identically in grayscale and in print, after Bringhurst's counsel that structure should be legible without ornament [1]. Every constant and equation is defined once, in a single macro file; every result is set at the grade it earned, and none was inflated in the typesetting.